

An Empirical Evaluation of LLMs for Microservice Bad Smell Detection: Results and Future Work

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Agenda

- ❑ Recapitulation
- ❑ Overall Detection Accuracy
- ❑ Detection Performance per Smell
- ❑ Error Analysis
- ❑ Discussion
- ❑ Future Work

Recapitulation

Evaluate 6 LLM models in detecting 5 types of Microservice Bad Smells (MBS) out of 9 Microservice Architectural Diagrams by comparison with a human-validated ground truth

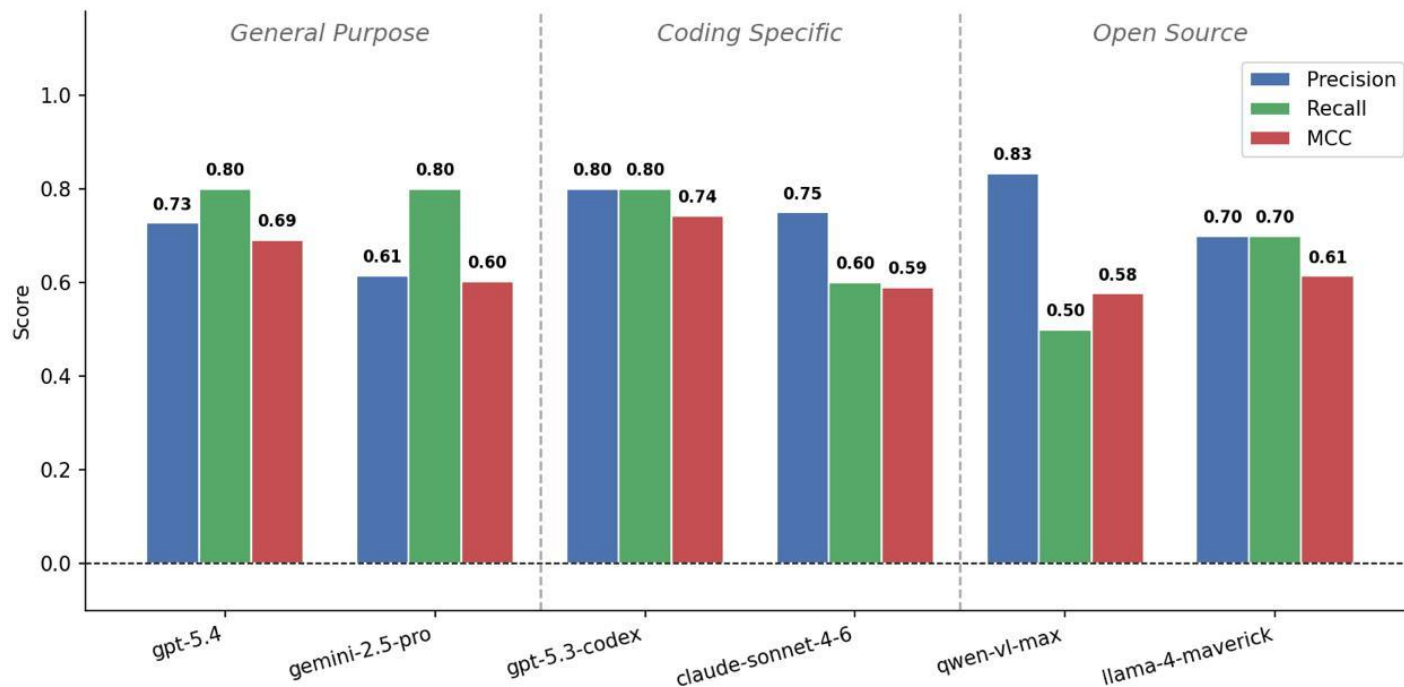
Category	Model	Provider
General purpose	GPT-5.4	OpenAI
	Gemini 2.5 Pro	Google
Coding specific	GPT-5.3-Codex	OpenAI
	Claude 4.6 Sonnet	Anthropic
Open-source	Qwen-vl-max	Alibaba
	Llama 4 Maverick	Meta

Types of MBS

- ☐ No API Gateway
- ☐ Nano Service
- ☐ Shared Database
- ☐ Microservice Greedy
- ☐ Cyclic Dependency

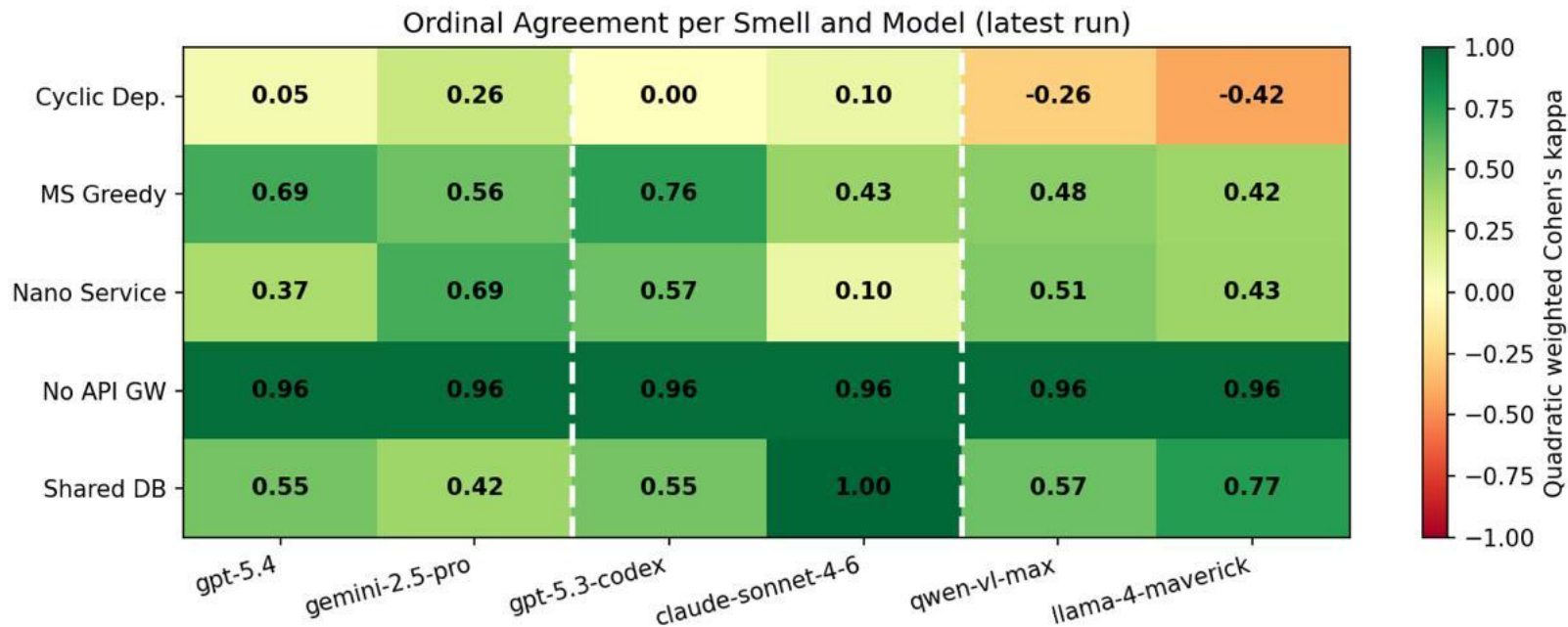
Overall Detection Accuracy

RQ1: To what extent can LLMs accurately Identify MBS from architectural diagrams?



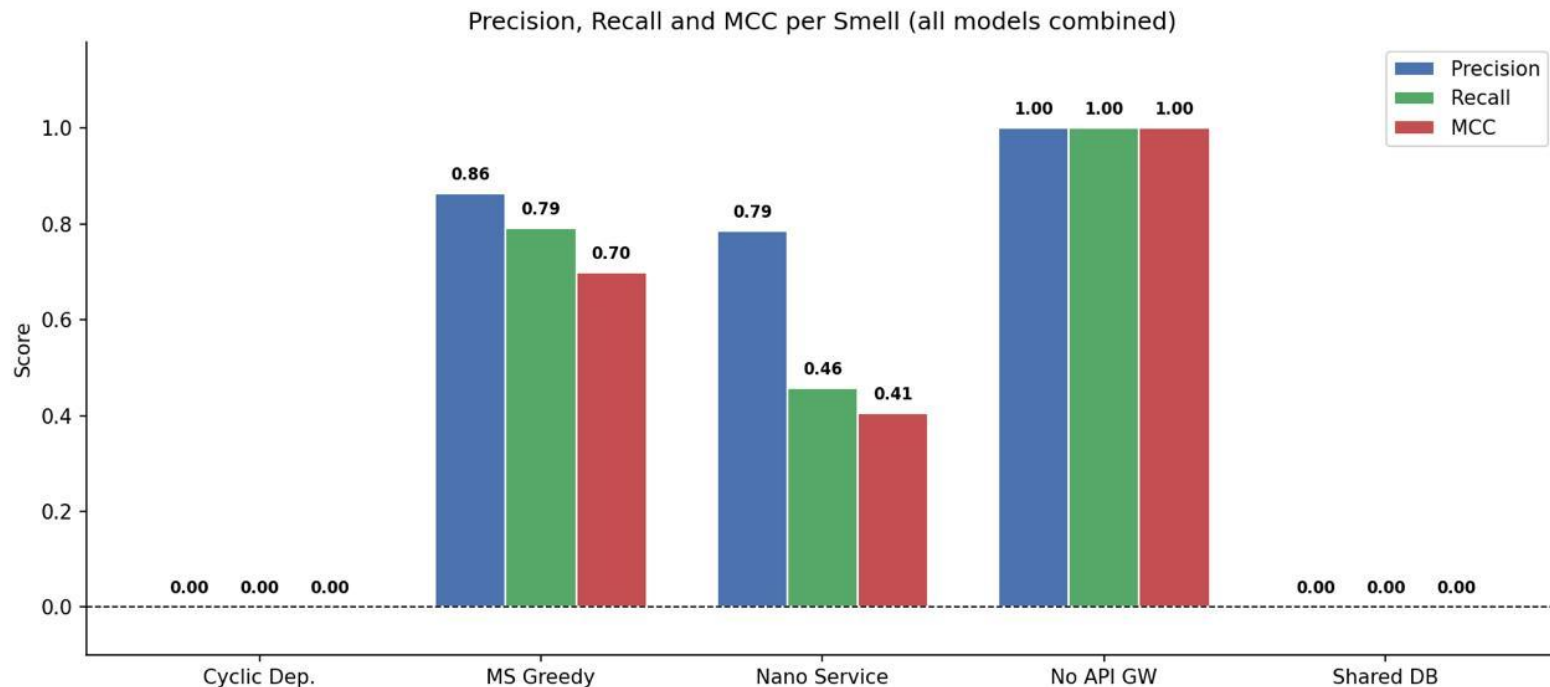
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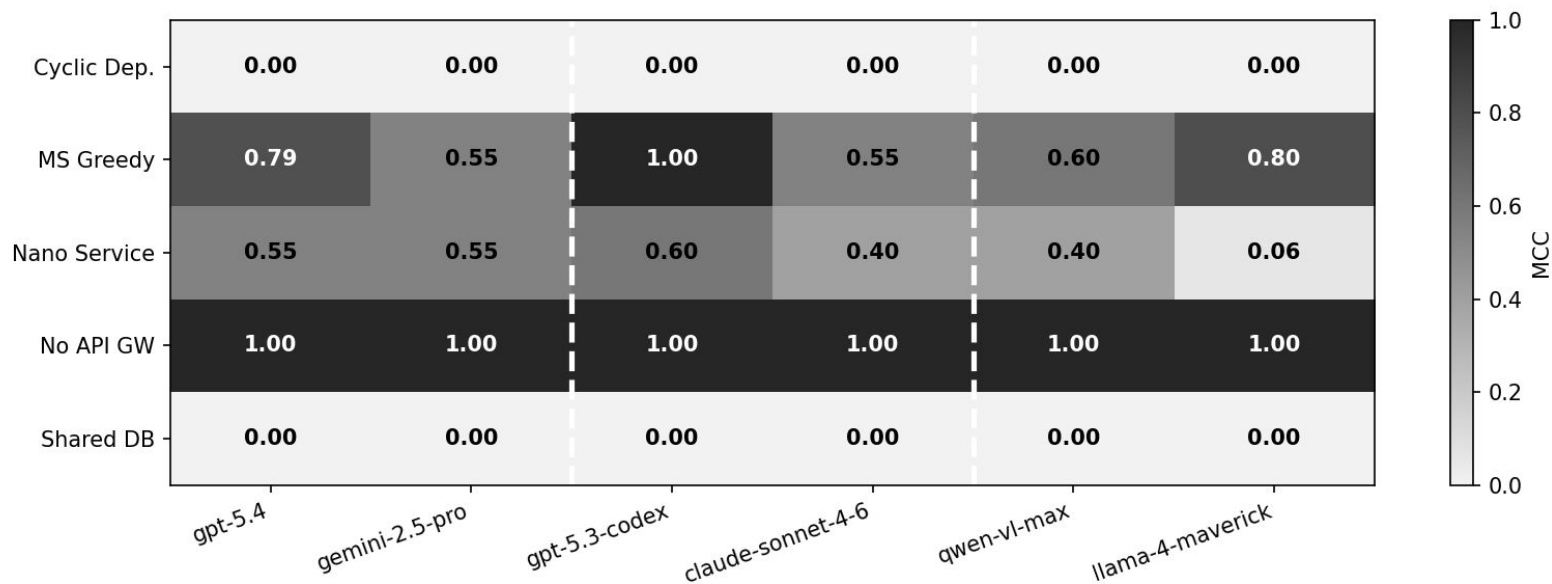
Detection Performance per Smell

RQ2: How does the detection performance of LLMs vary across the selected MBS?



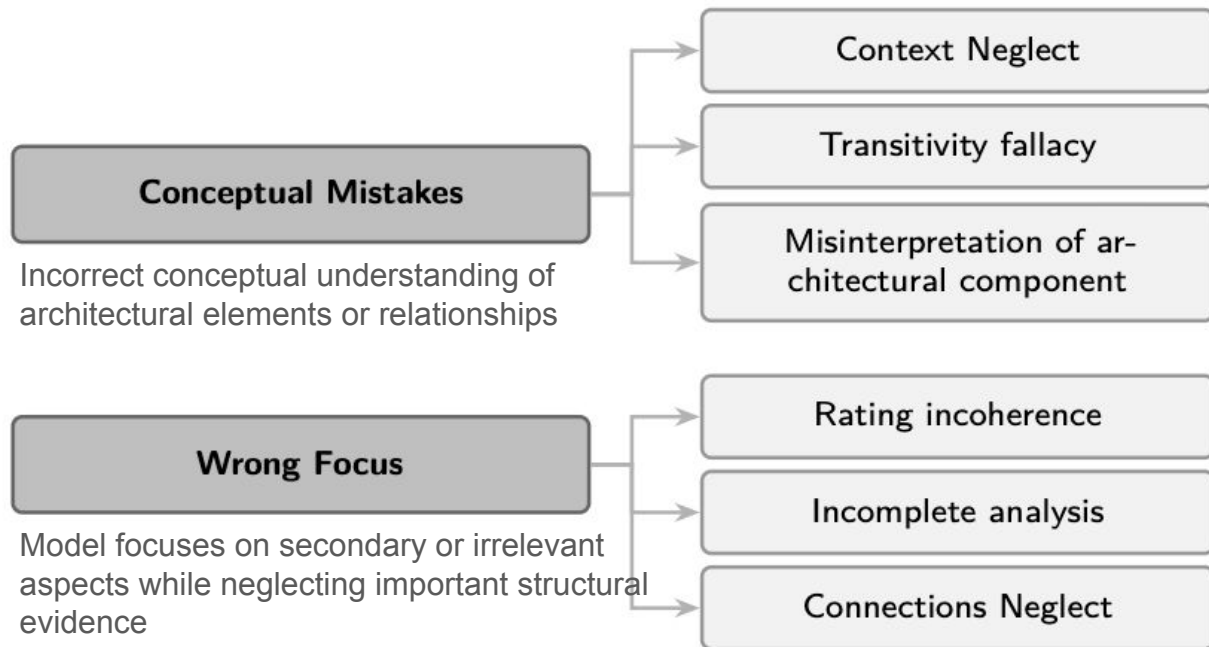
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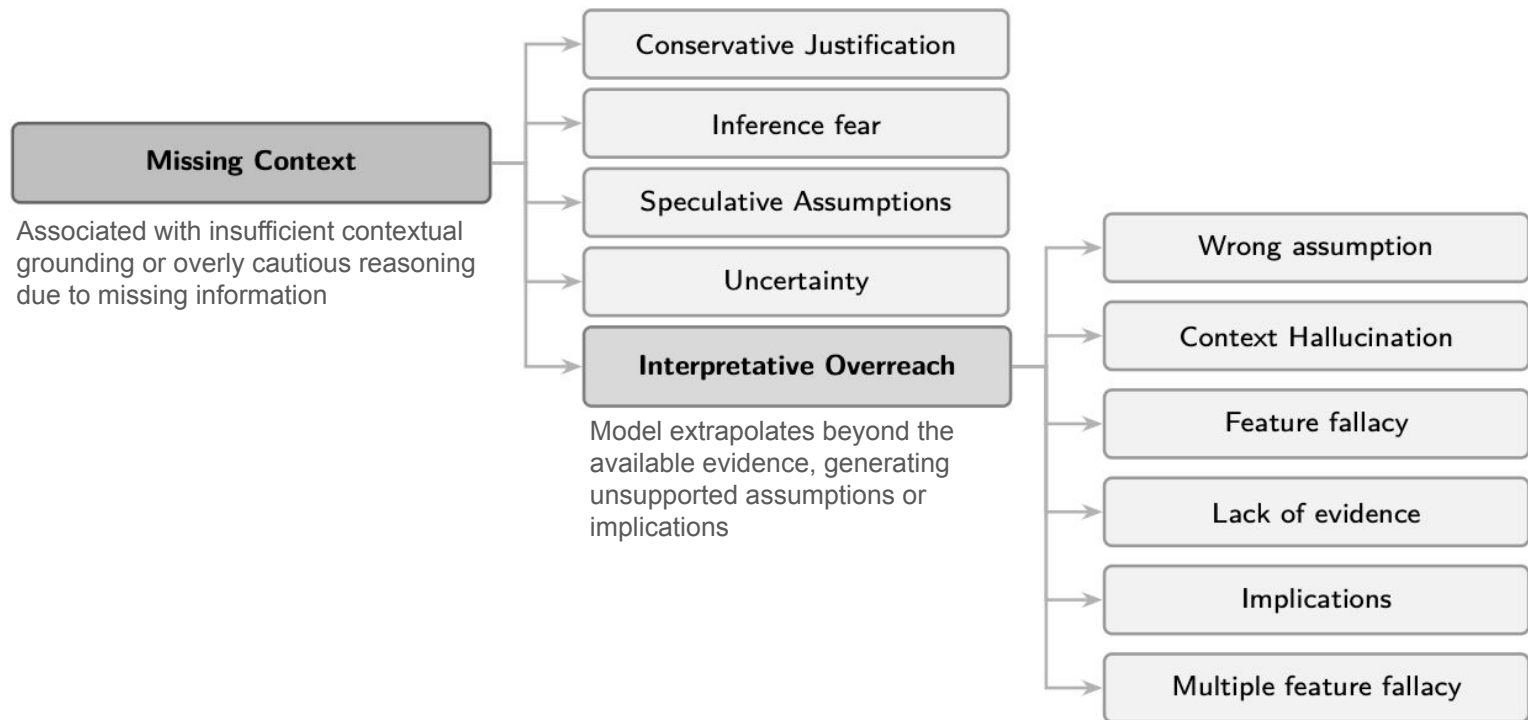
Error Analysis

RQ3: What are the primary root causes of errors made by LLMs during MBS detection?



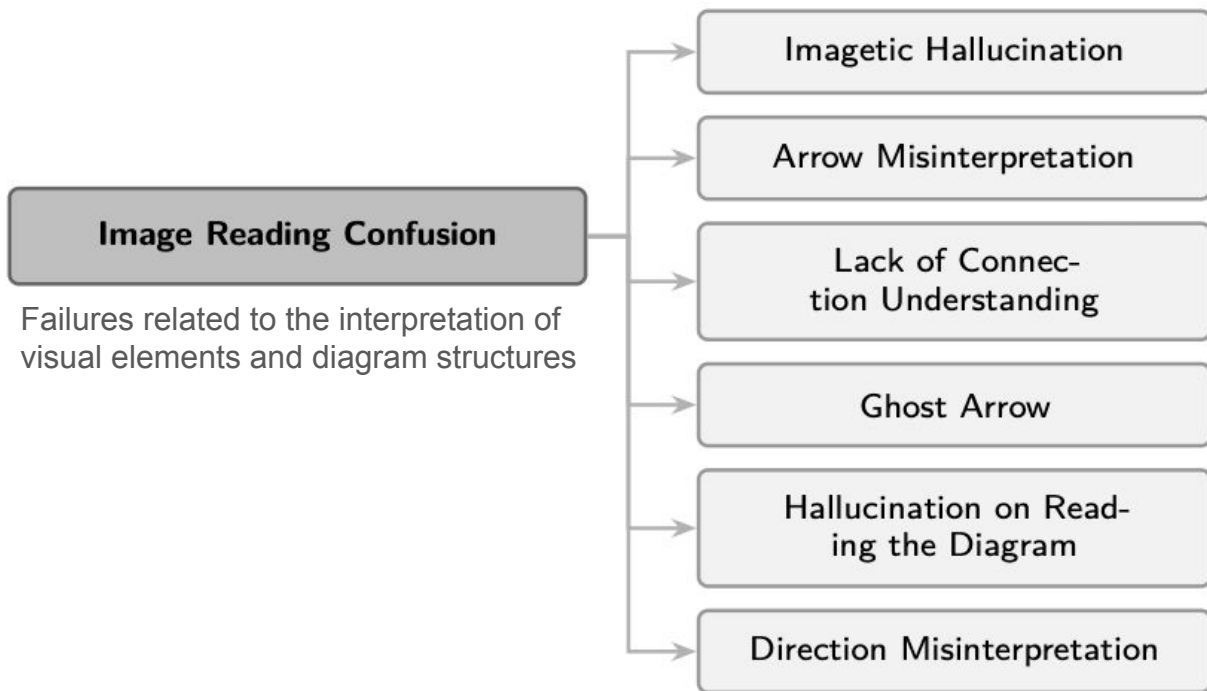
Error Analysis

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Discussion

- ❑ **Model specialization matters, but category alone is insufficient:** Coding-specific models achieved highest MCC (0.621) and precision (0.778), yet within-category variance (e.g., 0.144 MCC gap between GPT-5.3-Codex and Claude-Sonnet-4.6) exceeds cross-category averages.
- ❑ **Structural vs. semantic asymmetry:** Structural smells (e.g., missing API Gateway) are detected accurately, but complex patterns (e.g., Cyclic Dependency) suffer false positives; semantic smells yield high precision but low recall due to missing behavioral context.
- ❑ **Practitioner guidance:** Use code-specific multimodal LLMs as assistants with human-in-the-loop; augment diagrams with metadata to reduce hallucinations; prioritize early structural flaw detection.
- ❑ **Research directions:** Move beyond standalone visuals to multi-modal inputs (code, deployment descriptors); develop diagram pre-processing to reduce visual noise; create architecture-specific benchmarks and fine-tuning datasets.

Future Work

Idea 1: Evaluate LLMs for Microservice Architectural Diagram Refactoring (Removing MBS)

- ❑ Ask LLMs to refactor the diagrams that contain MBS
- ❑ Expert evaluation of AI Refactored Diagrams using the same criteria (presence of MBS)
- ❑ RQ1: To what extent are LLMs able to remove MBS from architectural diagrams?
- ❑ RQ2: Do LLMs introduce new MBS when refactoring architectural diagrams?
- ❑ RQ3: How the performance varies across the different models and MBS?

Future Work

Idea 2: Evaluate LLMs Capabilities when Generating Microservice Architectural Diagrams

- ❑ Ask LLMs to generate an architectural diagram based on the available repositories (description and source code) without the diagram
- ❑ Expert evaluation of all the diagrams (both AI and humans generated), then we perform a comparison between the AI vs human generated
- ❑ RQ1: To what extent are LLMs able to generate accurate architectural diagrams?
- ❑ RQ2: How do architectural diagrams generated by LLMs compare to human created ones?
- ❑ RQ3: What are the common problems with LLM-generated diagrams?

Future Work

Idea 3: Evaluate LLMs Capabilities when Describing Microservice Systems Based on its Architectural Diagram

- ❑ Ask LLMs to describe a microservice given an architectural diagram as input
- ❑ Expert evaluation of the descriptions, as well as comparison with descriptions available in the repositories
- ❑ RQ1: To what extent are LLMs able to describe a microservice system accurately based on its architectural diagram?
- ❑ RQ2: How the descriptions generated by AI compare to human descriptions?
- ❑ RQ3: What are the common mistakes AI make when describing the architectural diagrams?

Thank you!